

HOLY CROSS, AK, USA

WMO: 703051

Lat: 62.183N

Lon: 159.767W

Elev: 20

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26521

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.1	-27.0	-33.1	0.8	-27.1	-28.7	1.1	-21.9	21.6	6.8	19.7	10.7	0.9	250	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	75.8	59.1	71.9	58.3	67.9	56.4	61.4	71.7	59.5	68.9	57.8	65.6	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.1	63.5	55.5	65.7	62.0	54.1	62.2	60.6	27.3	71.4	26.0	68.7	24.9	65.8	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.3	14.1	-37.0	83.0	8.5	3.4	-43.1	85.5	-48.0	87.4	-52.7	89.3	-58.9	91.8		
(5)			-31.1	64.7	4.8	2.5	-34.6	66.5	-37.4	68.0	-40.1	69.4	-43.7	71.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.6	5.1	9.6	11.3	30.3	45.9	54.4	56.9	54.4	46.2	31.8	12.5	7.5		
	DBStd	22.89	19.99	16.00	12.57	11.49	8.59	6.05	5.59	5.00	6.43	8.76	14.26	15.06		
	HDD50	7685	1392	1130	1199	592	177	20	7	19	143	565	1124	1317		
	HDD65	12569	1857	1550	1664	1042	593	325	260	327	565	1030	1574	1782		
	CDD50	608	0	0	0	0	50	153	220	157	28	0	0	0		
	CDD65	16	0	0	0	0	1	7	8	0	0	0	0	0		
	CDH74	200	0	0	0	0	17	92	88	3	0	0	0	0		
	CDH80	33	0	0	0	0	1	18	14	0	0	0	0	0		
	WSAvg	5.6	6.2	7.5	6.9	5.8	5.3	4.8	4.7	4.3	4.8	5.1	5.9	5.6		
	PrecAvg	23.8	1.6	1.5	1.3	1.1	1.4	1.9	2.6	3.3	3.0	2.2	2.0	1.8		
(15) Precipitation	PrecMax	30.8	4.1	3.5	4.0	2.1	3.0	3.1	4.6	5.0	5.3	4.7	4.8	3.7		
	PrecMin	18.1	0.2	0.2	0.1	0.2	0.4	0.9	1.1	1.5	0.6	0.3	0.2	0.6		
	PrecStd	3.9	0.9	0.8	0.9	0.5	0.6	0.5	0.7	1.0	1.2	1.0	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	41.3	39.5	57.9	76.2	82.5	82.4	73.4	66.8	52.8	43.3	43.5		
		MCWB	37.6	36.8	34.4	44.5	56.7	61.3	62.8	59.6	54.9	46.7	38.6	37.8		
	2%	DB	39.6	37.8	36.5	53.6	71.2	76.3	75.7	69.0	62.1	48.4	38.1	38.6		
		MCWB	35.5	33.9	32.4	42.5	54.5	58.6	60.7	58.3	53.0	45.0	36.1	34.8		
	5%	DB	36.8	34.9	33.2	49.6	65.2	71.0	72.0	66.3	58.2	46.0	35.0	34.3		
		MCWB	33.7	32.1	30.1	40.5	50.9	56.5	59.7	56.9	51.2	43.1	33.5	32.5		
	10%	DB	32.8	31.5	29.8	45.9	60.1	66.4	67.7	63.6	55.3	43.7	31.9	29.4		
		MCWB	30.5	29.6	27.0	38.4	48.3	54.3	58.2	55.9	50.3	41.0	30.8	28.1		
	0.4%	WB	38.2	37.1	35.2	45.2	57.4	62.9	65.5	62.4	56.3	48.5	39.4	38.0		
		MCDB	42.6	40.6	37.9	56.0	74.5	78.4	76.7	68.8	63.2	51.4	40.7	40.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	34.5	33.1	42.8	54.8	60.0	62.4	59.9	53.9	45.6	36.2	35.0		
		MCDB	39.3	37.5	35.8	52.7	69.4	73.1	73.6	66.4	59.9	48.2	38.3	38.3		
	5%	WB	34.0	32.5	30.4	40.9	51.5	57.4	60.3	58.2	52.2	43.6	33.6	32.5		
		MCDB	37.0	34.7	33.4	49.2	63.2	69.2	69.9	64.1	57.2	46.0	35.0	34.6		
	10%	WB	30.8	29.5	27.0	38.8	49.1	55.0	58.6	56.9	50.8	41.3	30.7	28.4		
		MCDB	33.4	31.7	29.9	45.6	59.1	64.8	67.1	62.1	55.0	43.7	32.0	29.9		
	5% DB	MDBR	10.8	12.7	16.4	16.6	17.8	16.2	14.3	13.8	12.8	8.7	10.1	10.8		
		MCDBR	13.1	11.6	15.2	20.4	25.4	24.6	23.2	19.0	18.1	10.2	8.6	11.8		
		MCWBR	10.5	9.8	12.4	11.7	12.3	11.0	11.0	9.9	10.8	7.4	7.9	9.8		
		MCWBR	13.6	11.6	14.6	19.6	23.8	22.9	21.3	16.5	15.7	9.6	8.3	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.215	0.263	0.272	0.346	0.365	0.365	0.357	0.371	0.320	0.288	0.221	0.192		
		taud	2.036	2.229	2.268	2.131	2.330	2.384	2.451	2.416	2.535	2.431	2.053	1.906		
	Ebn at Noon		150	225	269	264	270	272	272	257	253	217	141	89		
		Edh at Noon	11	20	27	39	35	34	31	30	22	16	10	7		
(44) All-Sky Solar Radiation	RadAvg		95	317	906	1455	1653	1686	1420	1085	735	338	117	43		
	RadStd		13	38	99	126	174	158	117	115	86	34	18	5		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page